

Tokenization using NLTK

Program 1: Basic Word Tokenization using NLTK

Objective

Learn how to split text into individual words.

Install

```
pip install nltk
```

Python Code

```
import nltk
from nltk.tokenize import word_tokenize

# Download tokenizer data (run only once)
nltk.download('punkt')

text = "Artificial Intelligence is transforming the world!"

tokens = word_tokenize(text)

print("Original Text:")
print(text)

print("\nWord Tokens:")
print(tokens)

print("\nNumber of Tokens:", len(tokens))
```

Sample Output

```
Original Text:
Artificial Intelligence is transforming the world!
```

```
Word Tokens:
['Artificial', 'Intelligence', 'is', 'transforming', 'the', 'world', '!']
```

```
Number of Tokens: 7
```

Learning Outcome

- What is a token?
- Difference between text and tokens
- Punctuation becomes separate tokens.

Program 2: Sentence Tokenization

Objective

Split a paragraph into sentences.

```
import nltk
from nltk.tokenize import sent_tokenize

nltk.download('punkt')

paragraph = """
Artificial Intelligence is growing rapidly.
Machine Learning is a subset of AI.
Deep Learning uses neural networks.
Students enjoy learning NLP.
"""

sentences = sent_tokenize(paragraph)

print("Paragraph:\n")
print(paragraph)

print("\nSentences:\n")

for i, sentence in enumerate(sentences, start=1):
    print(f"Sentence {i}: {sentence}")
```

Sample Output

Sentence 1: Artificial Intelligence is growing rapidly.
Sentence 2: Machine Learning is a subset of AI.
Sentence 3: Deep Learning uses neural networks.
Sentence 4: Students enjoy learning NLP.

Learning Outcome

Students understand

- Sentence boundary detection
- Difference between sentence and word tokenization

Program 3: Tokenization with SpaCy

Objective

Perform professional NLP tokenization.

Install

```
pip install spacy  
python -m spacy download en_core_web_sm
```

Python Code

```
import spacy  
  
nlp = spacy.load("en_core_web_sm")  
  
text = "OpenAI develops powerful AI models like GPT."  
  
doc = nlp(text)  
  
print("Tokens:\n")  
  
for token in doc:  
    print(f"{token.text:15} -> {token.pos_}")
```

Sample Output

```
OpenAI      -> PROPN  
develops    -> VERB  
powerful    -> ADJ  
AI          -> PROPN  
models      -> NOUN  
like        -> ADP  
GPT         -> PROPN  
.           -> PUNCT
```

Learning Outcome

Students learn

- Industrial NLP library
- Token attributes
- Part-of-Speech tagging

Program 4: Hugging Face Tokenizer (LLM Tokenization)

Objective

Understand how GPT/BERT tokenize text into subwords.

Install

```
pip install transformers
```

Python Code

```
from transformers import AutoTokenizer

tokenizer = AutoTokenizer.from_pretrained("bert-base-uncased")

text = "Artificial Intelligence is amazing."

tokens = tokenizer.tokenize(text)

token_ids = tokenizer.convert_tokens_to_ids(tokens)

print("Original Text:")
print(text)

print("\nSubword Tokens:")
print(tokens)

print("\nToken IDs:")
print(token_ids)

print("\nDecoded Text:")
print(tokenizer.decode(token_ids))
```

Sample Output

Original Text:
Artificial Intelligence is amazing.

Subword Tokens:
['artificial', 'intelligence', 'is', 'amazing', '.']

Token IDs:
[7976, 4454, 2003, 6429, 1012]

Decoded Text:
artificial intelligence is amazing.

Learning Outcome

Students understand

- Modern LLM tokenization
- Token IDs
- Vocabulary mapping
- Subword tokenization